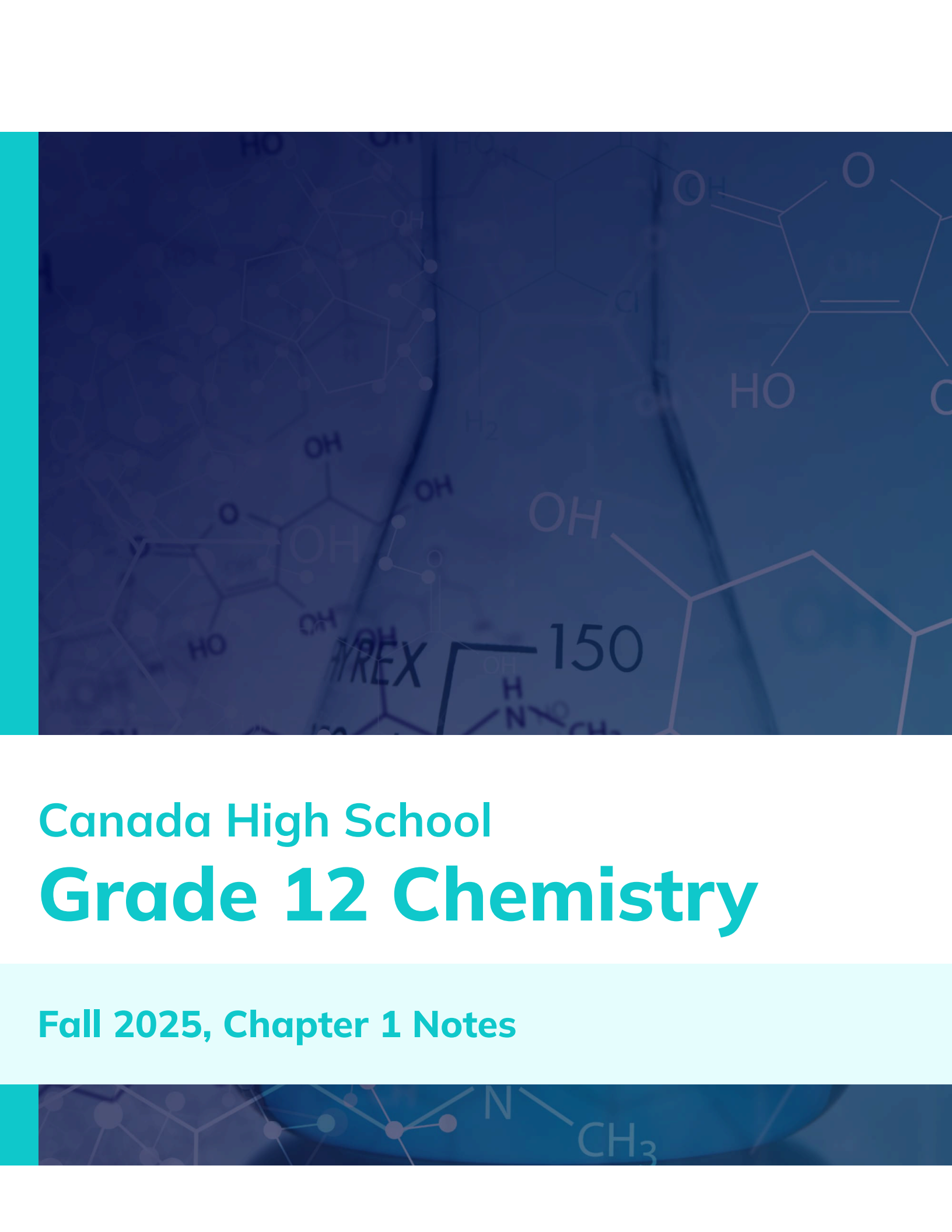




Canada High School Grade 12 Chemistry

Fall 2025, Chapter 1 Notes

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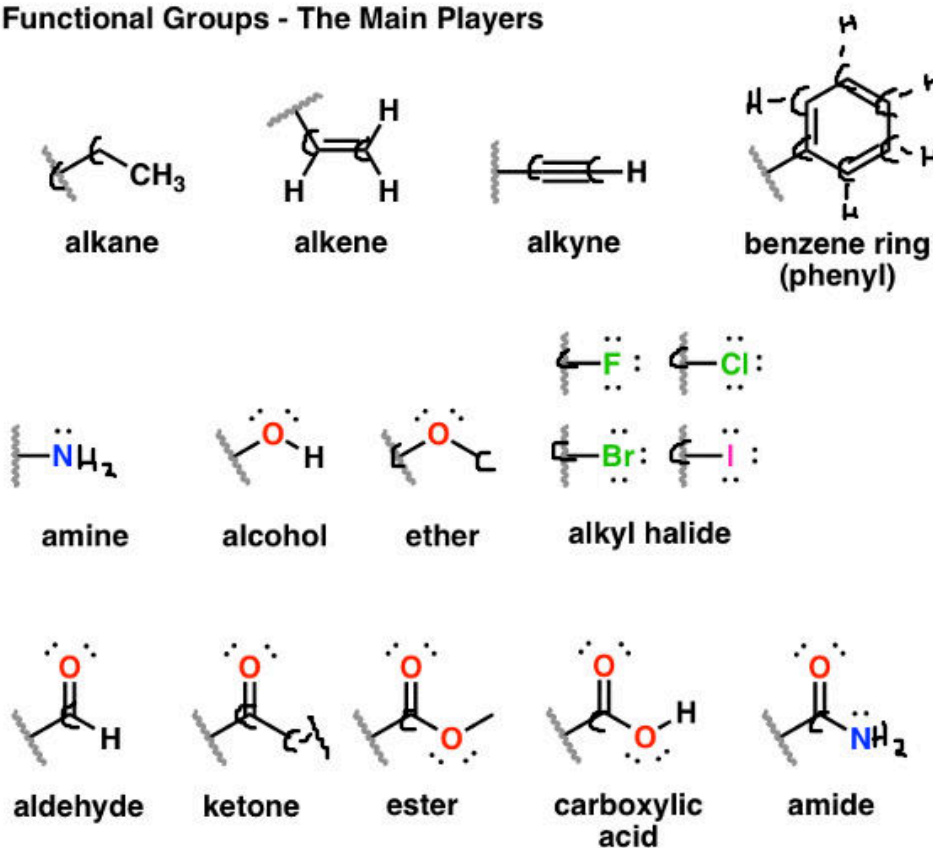
1. Organic Chemistry (Part 1: Functional Groups and Naming)

1.1 Hydrocarbons

1.1.1

Introduction to Functional Groups

Functional Groups - The Main Players



Functional groups determine whether:

- the molecules are soluble in polar or non-polar solvents
- the molecules have high or low melting and boiling points
- molecules readily react with other molecules

3 Main Components of Functional Groups

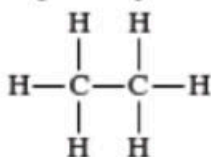
- There are many different functional groups but they essentially consist of only 3 main components:
 1. Carbon-Carbon multiple bonds (-C=C- or $\text{-C}\equiv\text{C-}$)
 2. Single bonds between a Carbon atom and a more electronegative atom (eg. C-O , C-N , or C-Cl)
 3. Carbon atom double bonded to an oxygen atom (-C=O)

Introduction to Hydrocarbons

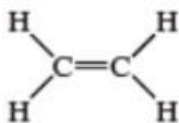
- A **hydrocarbon** is an organic compound that only contains **carbon** and **hydrogen** atoms

Structures of representative hydrocarbons

aliphatic hydrocarbons



alkane

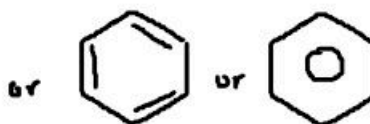
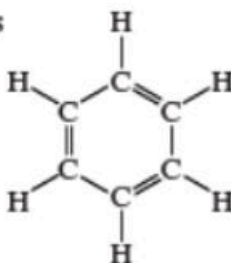


alkene



alkyne

aromatic hydrocarbons



Unsaturated vs Saturated Hydrocarbons

- A **saturated hydrocarbon** is one that has the maximum number of hydrogens added (no double or triple bonds)
 - Alkanes are saturated hydrocarbons
 - While alkenes and alkynes are unsaturated hydrocarbons

General Formulas for Hydrocarbons

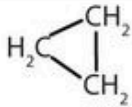
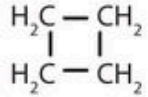
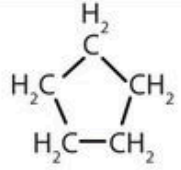
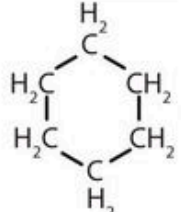
- Alkanes** are hydrocarbons that only have single bonds between carbon atoms ... $\text{C}_n\text{H}_{2n+2}$
- Alkenes** are hydrocarbons that have at least 1 C=C double bond ... C_nH_{2n}
- Alkynes** are hydrocarbons that contain at least 1 C≡C triple bond ... $\text{C}_n\text{H}_{2n-2}$

Aliphatic vs Aromatic Hydrocarbons

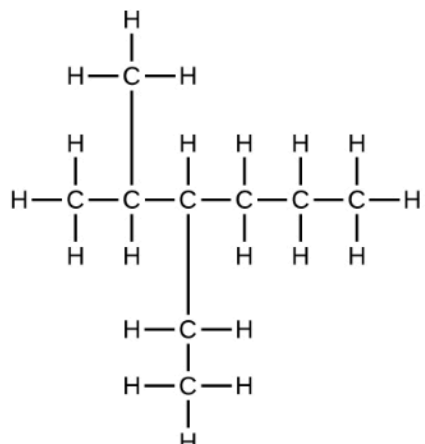
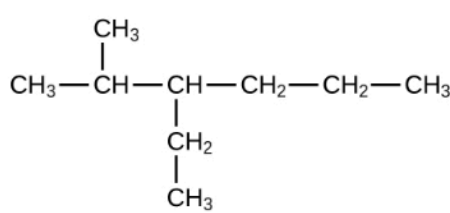
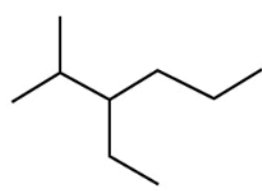
- Aliphatic hydrocarbons:** compounds that have a structure based on straight or branched chains or rings of carbon atoms (does not include aromatic compounds like benzene)

-
- Includes: alkanes, alkenes, alkynes, and cyclic hydrocarbons
 - **Aromatic hydrocarbons:** a compound with a structure based on benzene (C_6H_6).
 - benzene has all 6 Cs in intermediate length between a single and double bond because of resonance

Cyclic Hydrocarbons

Name	Molecular Formula	Structural Formula
cyclopropane	C_3H_6	 or 
cyclobutane	C_4H_8	 or 
cyclopentane	C_5H_{10}	 or 
cyclohexane	C_6H_{12}	 or 

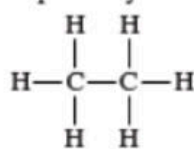
5 Ways to Represent Alkane Structures

 Structural Formula (shows bonds between each atom)	 Condensed Formula (doesn't show bonds between C and H)	 Line Structural Formula (end points and intersections represent 1 C atom, Hydrogens are assumed to be present)
C_9H_{20} Empirical Molecular Formula	$CH_3CH(CH_3)CH(CH_2CH_3)CH_2CH_2CH_3$ Expanded Molecular Formula	

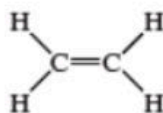
Properties of Alkanes, Alkenes, & Alkynes

Structures of representative hydrocarbons

aliphatic hydrocarbons



alkane

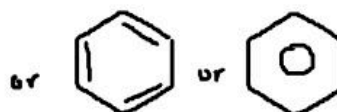
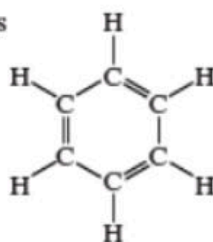


alkene



alkyne

aromatic hydrocarbons



- Are non-polar
- Soluble in non-polar solvents (Like dissolves like)
- Don't dissolve in water (polar)
- **Boiling point increases as the chain gets longer** due to increased intermolecular forces (London dispersion forces)
- **Highly branched alkanes have lower boiling points**
 - *Think: tree trunk vs small branches
- **Alkenes have lower boiling points than alkanes** with the same number of carbon atoms
- **Alkynes have higher boiling points than alkanes** with the same number of carbon atoms (it has to do with the triple bond and how it creates a linear structure, which causes the atoms to attract each other more strongly, so it takes more energy to break them apart)
- Aromatic hydrocarbons have similar boiling points as aliphatic hydrocarbons with the same # of carbons
- Benzene is a liquid at room temperature

-
- Aromatic compounds were named this because they are "aromatic" and often have strong odours

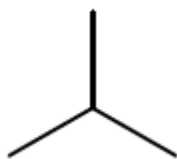
Boiling Points of the First 10 Straight Alkanes

Alkane	Molecular Formula	Melting Point (°C)	Boiling Point (°C)
methane	CH ₄	-182.5	-161.5
ethane	C ₂ H ₆	-183.3	-88.6
propane	C ₃ H ₈	-187.7	-42.1
butane	C ₄ H ₁₀	-138.3	-0.5
pentane	C ₅ H ₁₂	-129.7	36.1
hexane	C ₆ H ₁₄	-95.3	68.7
heptane	C ₇ H ₁₆	-90.6	98.4
octane	C ₈ H ₁₈	-56.8	125.7
nonane	C ₉ H ₂₀	-53.6	150.8
decane	C ₁₀ H ₂₂	-29.7	174.0

1.1.6

For the following pair of molecules, determine which will have the higher melting point.

Molecule 1:



VS

Molecule 2:



1.2 Naming Alkanes

1.2.1

Naming Alkanes

- For alkanes the suffix (ending) is **-ane**
- The root of the name indicates the number of carbon atoms in the longest straight chain in the molecule
- IUPAC naming will always follow the general form: **prefix + root + suffix**
 - **prefix**=identifies side chains
 - **root**=identifies the number of carbon atoms in the main/parent chain
 - **suffix**=indicates the family of compounds to which it belongs

carbon atoms = root

1 = meth-

2 = eth-

3 = prop-

4 = but-

5 = pent-

6 = hex-

7 = hept-

8 = oct-

9 = non-

10 = dec-

i WIZE TIP

You can remember the roots with the following sentence:

Mary Eats Peanut Butter Pancakes Happily Hugging Our New Dog

Ex.



We can also see branches in the chain:

- An **alkyl group** is a hydrocarbon group derived from an alkane but has a hydrogen atom removed
 - These are often branches on organic molecules
 - ex. **Methane** is CH_4 , but when it is attached to the main chain of a molecule, we call this **branch a methyl group $-\text{CH}_3$**
 - ex. **Ethane** is CH_3CH_3 , but when it is attached to the main chain of a molecule, we call this **branch an ethyl group $-\text{CH}_2-\text{CH}_3$**

Alkanes and Related Alkyl Groups

Alkane	Name	Alkyl Group	Name
CH_3-H	Methane	CH_3-	Methyl
$\text{CH}_3\text{CH}_2-\text{H}$	Ethane	CH_3CH_2-	Ethyl
$\text{CH}_3\text{CH}_2\text{CH}_2-\text{H}$	Propane	$\text{CH}_3\text{CH}_2\text{CH}_2-$	Propyl
$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2-\text{H}$	Butane	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2-$	Butyl
$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2-\text{H}$	Pentane	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2-$	Pentyl

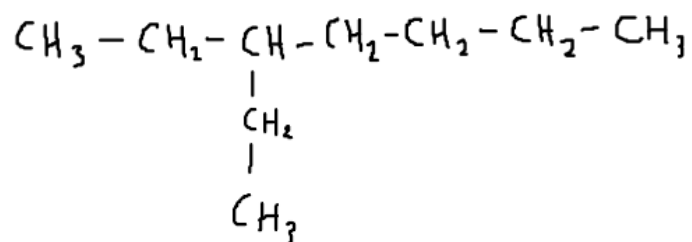
Rules for Naming Branched Alkanes

WIZE CONCEPT

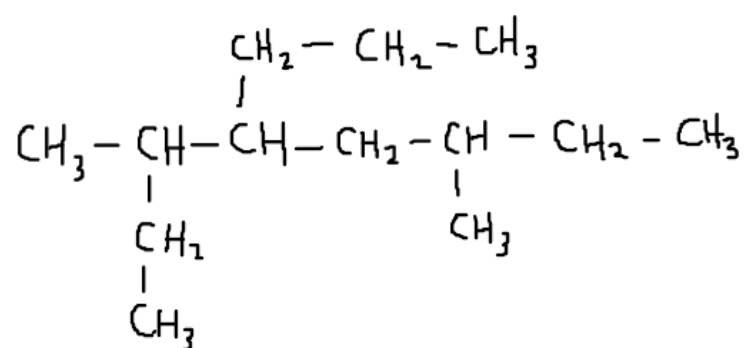
1. Identify the longest carbon chain
2. Number the carbon atoms in the longest carbon chain starting from the end that is closest to the branches (we do this because we want the branches to have the lowest numbers possible)
3. Name each branch and identify its location on the parent chain.
4. Write the IUPAC name in the form (# of location)-(branch name)(parent chain).
5. If there is more than 1 branch present, list the branches in alphabetical order. If there is more than one substituent of the same type, then we combine them with a prefix such as di, tri, tetra (these prefixes don't affect alphabetical order). Each number is separated by a comma (,).

Note: When naming cyclic hydrocarbons, if the carbon atoms that form the ring structure are the parent chain, the prefix cyclo is added to the parent hydrocarbon name, name the substituted groups/branches as you would normally. If the ring is not the parent chain, it will be labelled as a branch (ex. 3-cyclopentylheptane indicates a cyclopentyl ring attached to the 3rd carbon in a chain of 7 carbons).

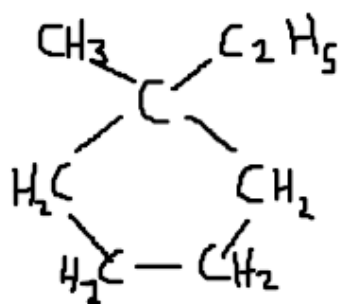
Ex. Name the following molecule:



Ex. Name the following molecule:

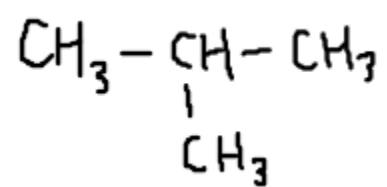


Ex. Name the following molecule:



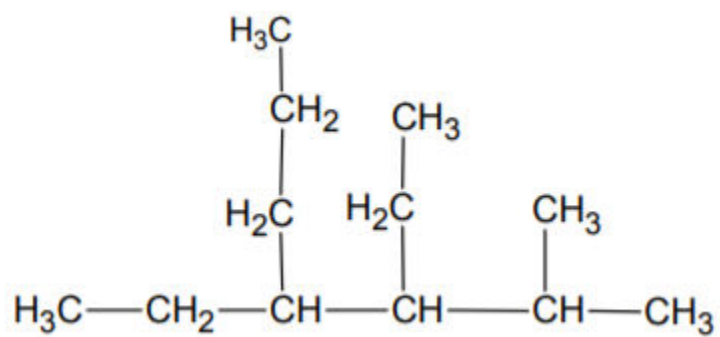
1.2.2

Name the following molecule:



1.2.3

Name the following molecule:



1.2.4

Draw the following molecules:

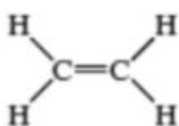
1. 3, 3, 5-trimethyloctane

2. 1, 1-diethylcyclohexane

1.3 Naming Alkenes and Alkynes

1.3.1

Naming Alkenes and Alkynes



alkene



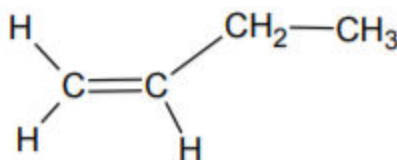
alkyne

WIZE CONCEPT

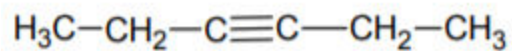
1. The parent chain must include the alkene or alkyne (must contain a multiple bond)
2. When numbering the longest chain, number it so that the multiple bond gets the lowest possible number.
3. To indicate where the double bond is in the parent chain, we write the prefix for the long chain-(# of C atom that begins the multiple bond)-suffix (either ene or yne depending on if it is an alkene or alkyne). For example, if the molecule has 5 carbons in a straight chain, and a double bond is between carbons 2 and 3, we write: pent-2-ene.

Note: For cycloalkenes and cycloalkynes, naming is the same as for cycloalkanes but the double or triple bond is given the 1-2 position (since we want it to have the lowest numbers) and the other substituents are numbered to give the lowest combination of numbers.

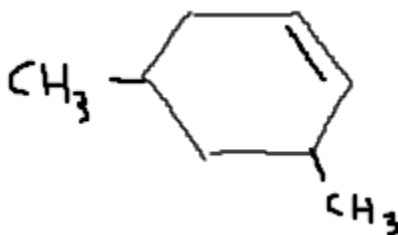
Example: Name the following molecule



Example: Name the following molecule

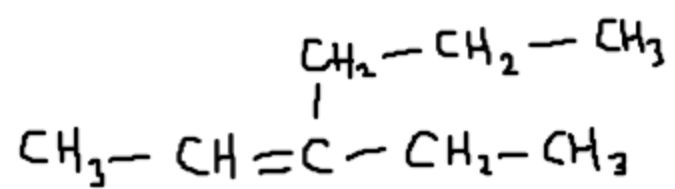


Example: Name the following molecule



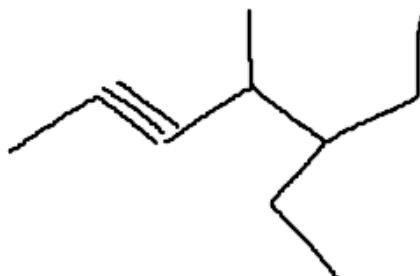
1.3.2

Name the following molecule:



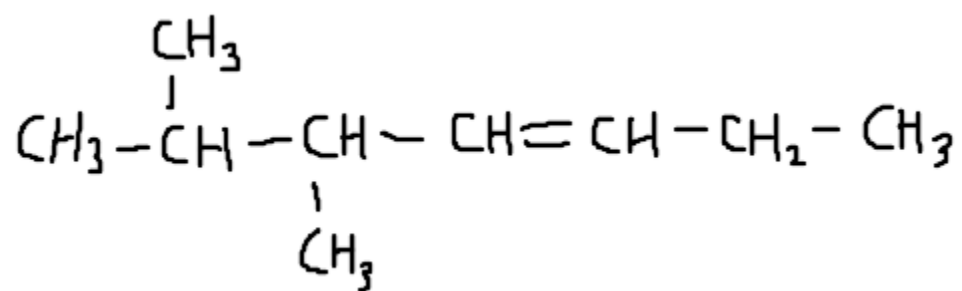
1.3.3

Name the following molecule:



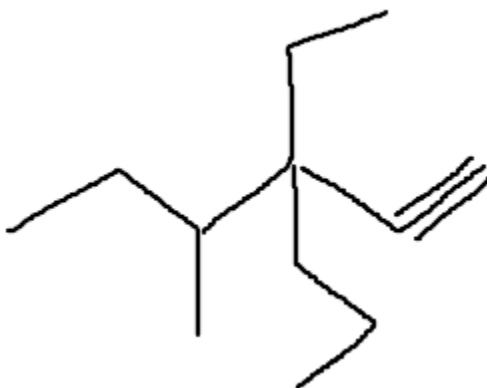
1.3.4

Name the following molecule:



1.3.5

Name the following molecule:

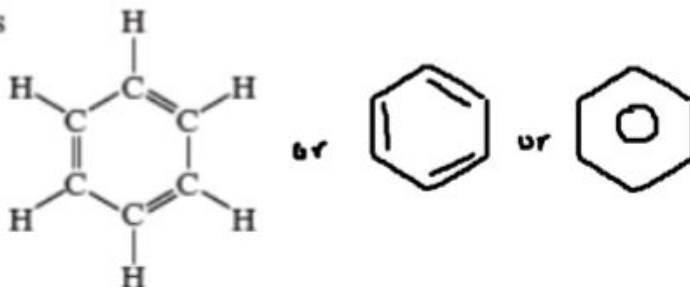


1.4 Naming Aromatic Hydrocarbons

1.4.1

Naming Aromatic Hydrocarbons

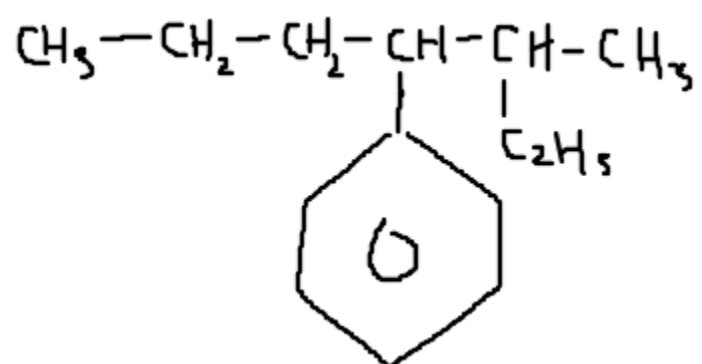
aromatic hydrocarbons



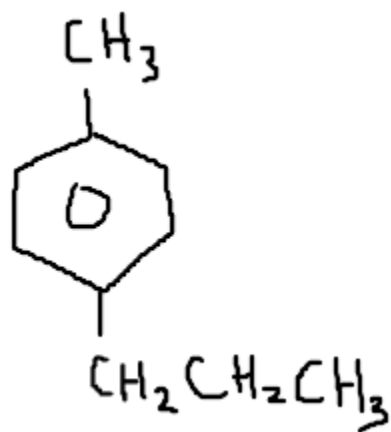
WIZE CONCEPT

1. Identify the root (the longest carbon chain). If the aromatic benzene ring is the root, we call it benzene, but if the aromatic benzene ring is a branch off of the parent chain, we call it phenyl.
2. If more than one group is attached to the benzene ring, the groups are numbered with the lowest possible numbers.
3. When there are different groups on the benzene ring, if possible, give the lowest number to the group that comes first in alphabetical order.

Example: Name the following molecule

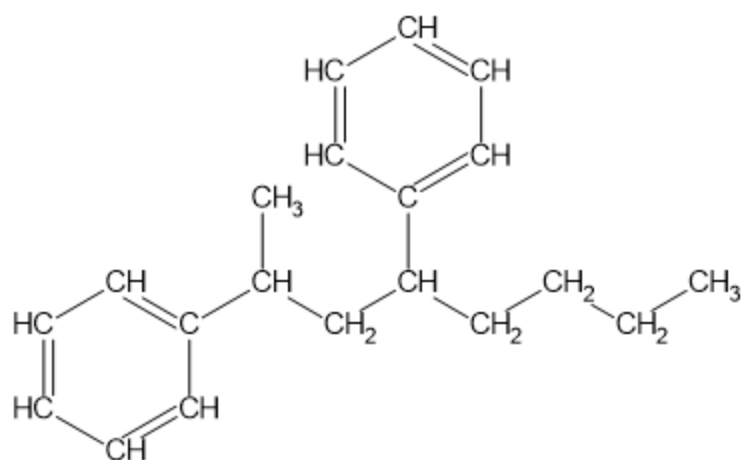


Example: Name the following molecule



1.4.2

Name the following molecule:



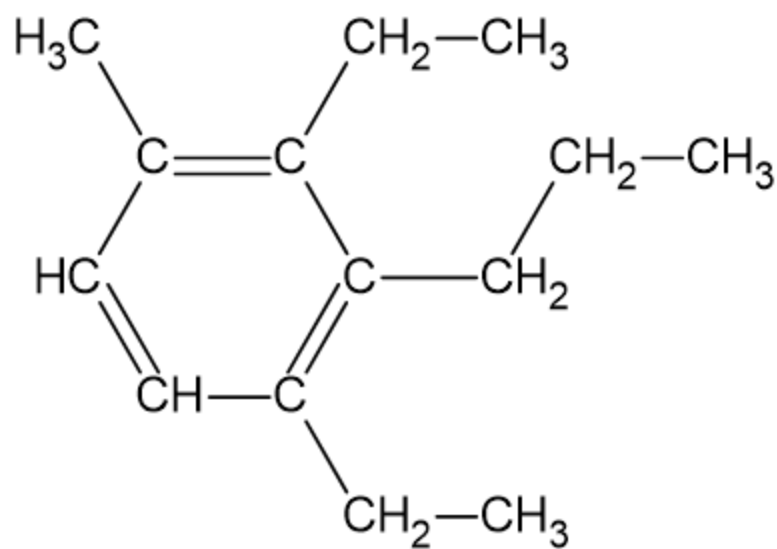
1.4.3

Draw the following molecule:

1-ethyl-3-propylbenzene

1.4.4

Name the following molecule

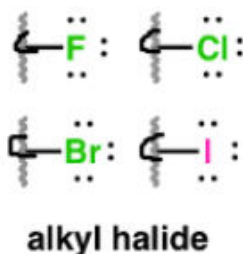


1.5 Naming Alkyl Halides

1.5.1

Naming and Properties of Alkyl Halides

- Alkyl halides (aka haloalkanes) have replaced a hydrogen with a halogen!



Properties:

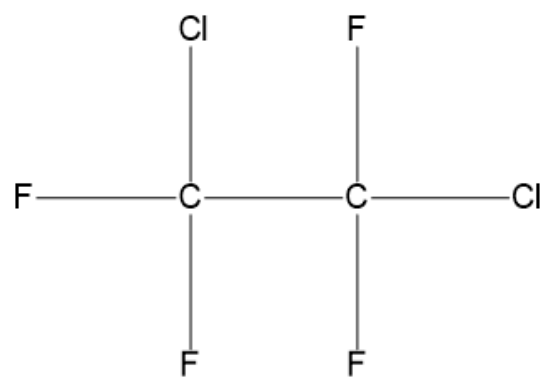
- More polar than corresponding parent alkanes -> stronger intermolecular forces -> higher melting and boiling points
- The smallest haloalkanes (ex. CH_3Br) are slightly soluble in water, the rest are insoluble

Naming:

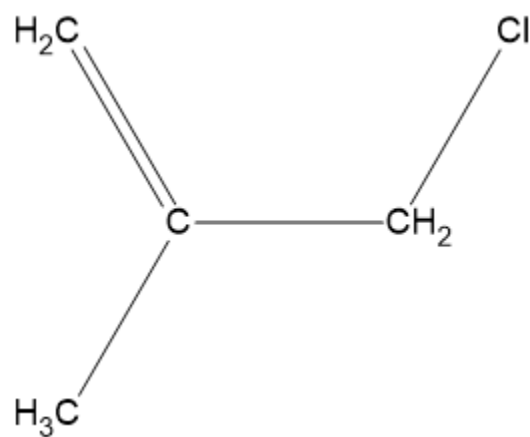
WIZE CONCEPT

- Name the parent hydrocarbon chain
- Identify the specific halogen involved with a prefix (F=fluoro, Cl=chloro, Br=bromo, I=iodo). Write the prefixes alphabetically along with any other groups present (ex. ethyl)
- Identify the total numbers of each halogen using prefixes (di, tri, tetra, etc)
- Indicate the position of the halogen in the chain (try to give the halogen low numbers when possible)

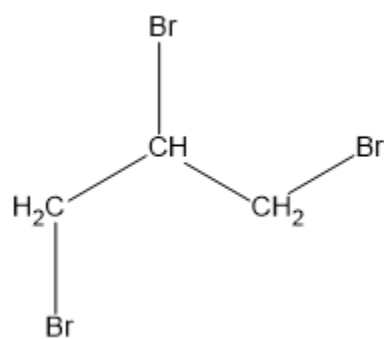
Example: Name the following



Example: Name the following



1.5.2



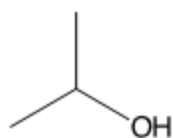
1.6 Isomers

1.6.1

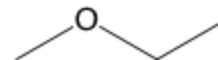
Isomerism

Isomers are compounds that have the same molecular formula but their atoms are in a different arrangement

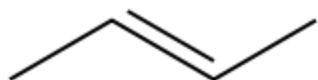
1. **Constitutional isomers** possess a different connectivity from one another. This means while the molecular formulae are the same, there can be many possibilities for different functional groups or bonding patterns.



and



-
- 2. **Stereoisomers** possess the same functional groups in the same connectivity (so they also have the same molecular formulae), but they have different 3D orientations of their atoms in space.
- **Configurational isomers** are a type of **stereoisomer** where the difference in spatial arrangement is **not due to rotation** or twisting about a single bond.
 - **2a) Diastereomers (cis-trans isomers)** are based on a double bond where different types of atoms or groups are bonded to each carbon atom in the double bond.



trans-but-2-ene

and



cis-but-2-ene

i WIZE TIP

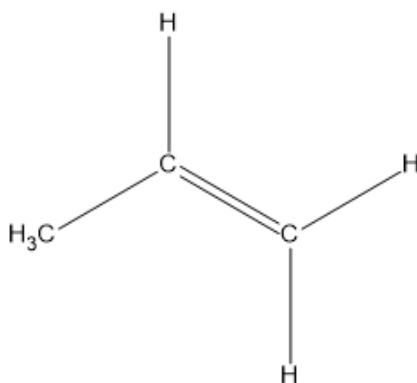
For the *cis* isomer: Both of the largest functional groups are on the same side

*Think cis are like sisters, so they are on the same side!

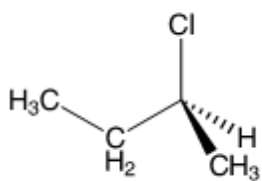
For the *trans* isomer: Both of the largest functional groups are on opposite sides.

! WATCH OUT!

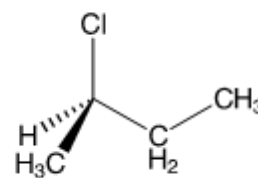
Diastereomers are not possible if two groups on one carbon are the same (see below for an example)



- **2b) Enantiomers** a type of stereoisomer where molecules are mirror images of each other around a single carbon atom that is connected to 4 different types of atoms or groups

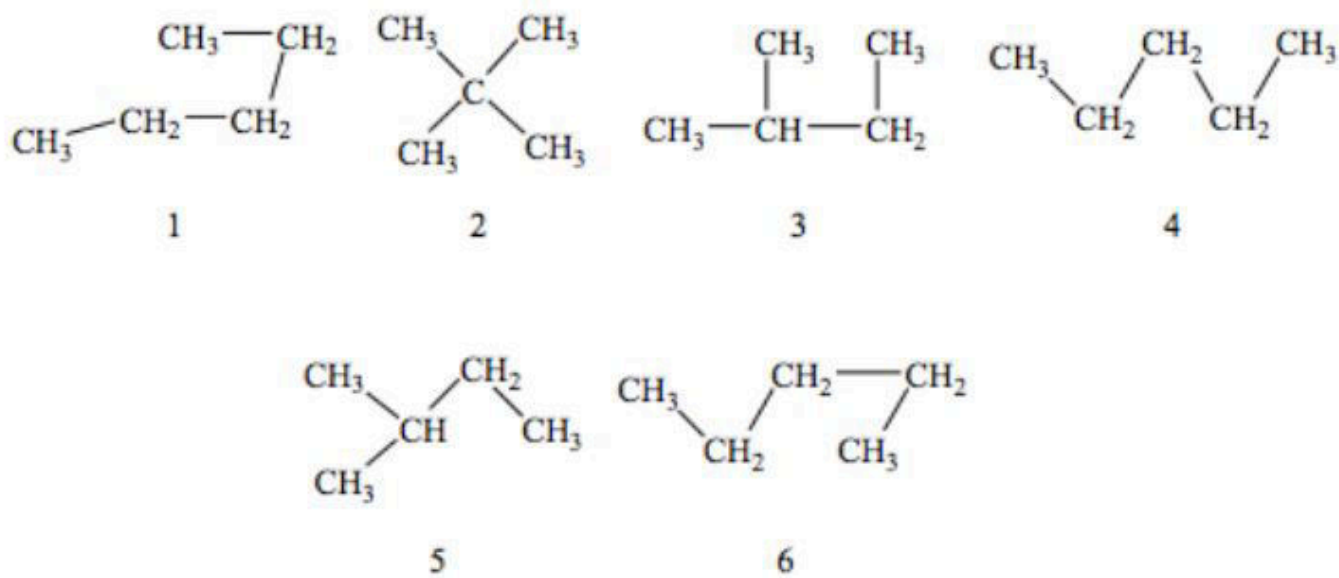


and



1.6.2

The alkane pentane, C_5H_{12} , has several structures (i.e. constitutional isomers, all of which are included in the set of molecules shown below.



Which of the following sets of structures presents the isomers of pentane without counting any isomer more than once?

1,2,3, and 4

☐

2,3, and 4

☐

2,4,5, and 6

☐

2, 3, and 5

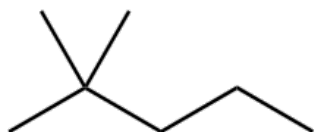
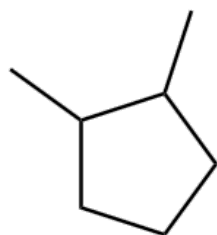
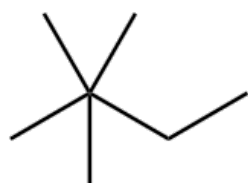
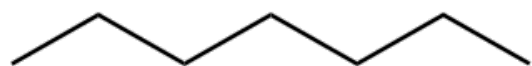
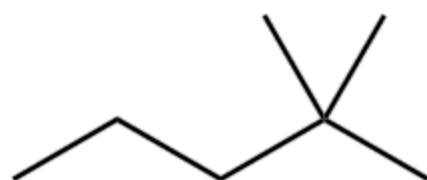
☐

2, 4, and 6

☐

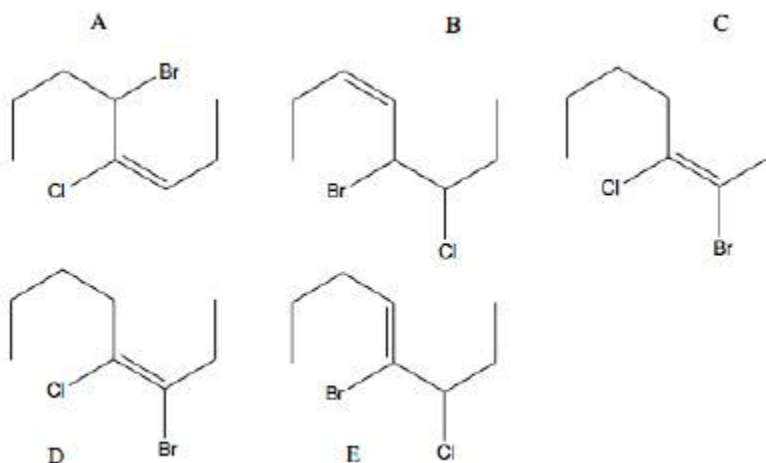
1.6.3

Choose the compound below which is a structural isomer of 2,2-dimethylpentane

☐☐☐☐☐

1.6.4

Which of the following drawings is cis – 5 – bromo – 6 – chlorooct-3-ene?



A

☐

B

☐

C what we can do.

☐

D

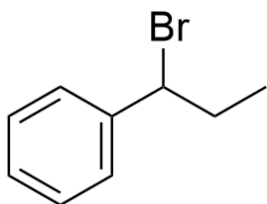
☐

E

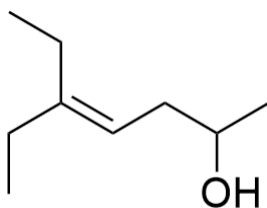
☐

1.6.5

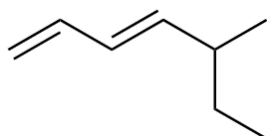
Which of the following molecules has both *cis/trans* isomers and has an enantiomer?



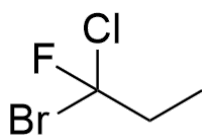
☐



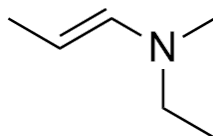
☐



☐



☐

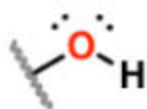


☐

1.7 Alcohols

1.7.1

Naming and Properties of Alcohols



alcohol

Properties:

- OH group is polar (can form hydrogen bonds)
- Boiling points are much higher than corresponding alkanes
- Smaller alcohols are a lot more polar, as the alcohols get longer (ex. pentanol), they are less polar

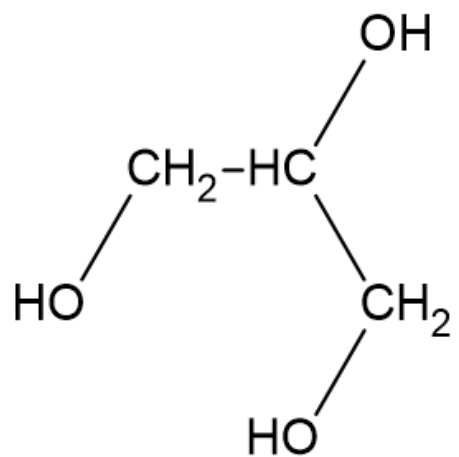
Naming:



WIZE CONCEPT

1. Identify the longest carbon chain.
2. The suffix of an alcohol is "ol" (ex. butan-2-ol)
3. If there is more than one OH group, place a prefix (di, tri, tetra) in front of the "ol" ending (ex. butane-1,2-diol)

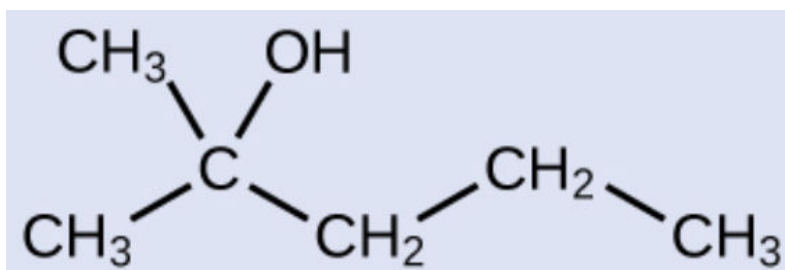
Example: Name the following molecule



Example: Draw the following molecule
4-ethyloctan-2,5-diol

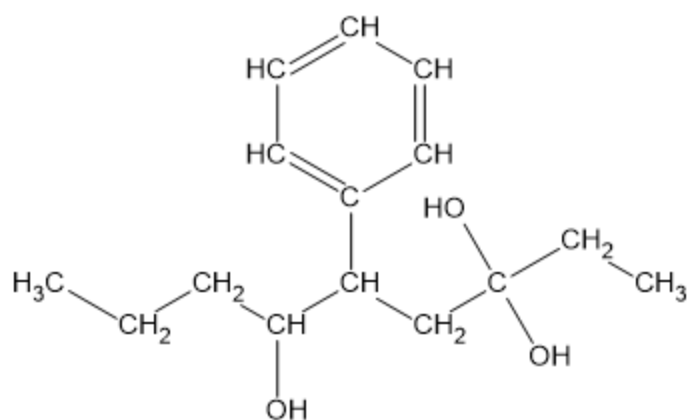
1.7.2

Name the following molecule:



1.7.3

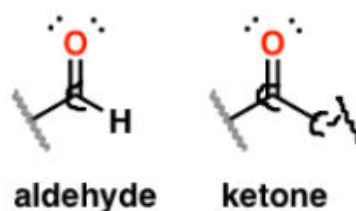
Name the following molecule



1.8 Aldehydes and Ketones

1.8.1

Naming and Properties of Aldehydes and Ketones



Properties:

- The **carbonyl group is very polar**
- These cannot hydrogen bond with themselves because they don't have OH bonds, but can hydrogen bond with water since they have a very electronegative O atom.
- Recall that alcohols have the OH group and are thus able to hydrogen bond with themselves and with water
- This is why **aldehydes and ketones have lower boiling points than analogous alcohols and they are less soluble in water than alcohols**
- But because of the polar carbonyl group, **they are more soluble in water than hydrocarbons are**

Naming:

WIZE CONCEPT

For aldehydes:

1. Take the parent alkane name, drop the final "e" and add the suffix "al" ex. pentanal.
2. The aldehyde group is always carbon 1.

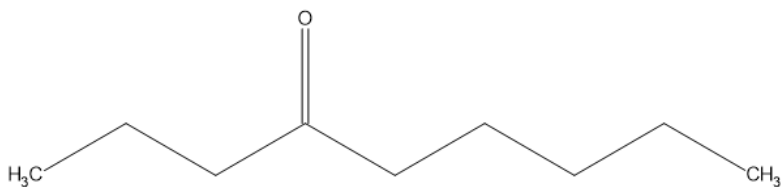
For ketones:

1. Take the parent alkane name, drop the final "e" and add the suffix "one".
2. Use a number to indicate the location of the carbonyl group. (ex. pentan-2-one) and give the carbonyl carbon the lowest numbers possible.

Example: Draw the following molecule

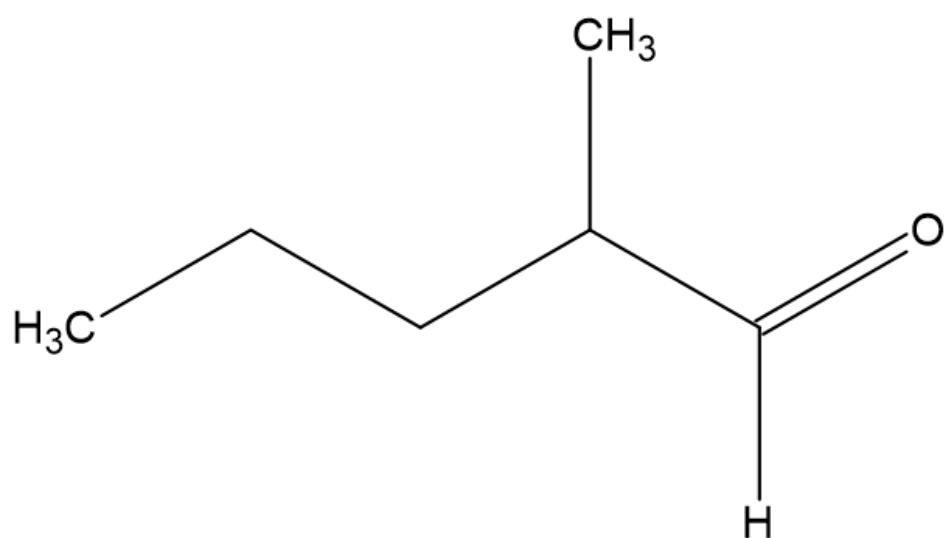
2-chloro-2-methylpentanal

Example: Name the following molecule



1.8.2

Name the following molecule



1.8.3

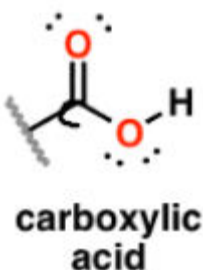
Draw the correct structural or line formula for the following molecule:

4,4-diethylheptane-2,3-dione

1.9 Carboxylic Acids

1.9.1

Naming And Properties of Carboxylic Acids



- These are characterized by a carboxyl group (-COOH)

Properties:

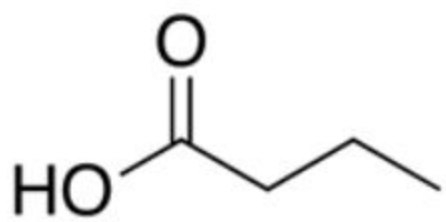
- The presence of both the carbonyl (-C=O) and the hydroxyl (-OH) groups makes the **carboxylic acid polar**
- They can also **form hydrogen bonds** with each other and water
- They have **higher boiling points than the corresponding hydrocarbons**
- smaller carboxylic acids are very soluble in water (1-4C) while chains with 10+ carbons are insoluble in water

Naming:

WIZE CONCEPT

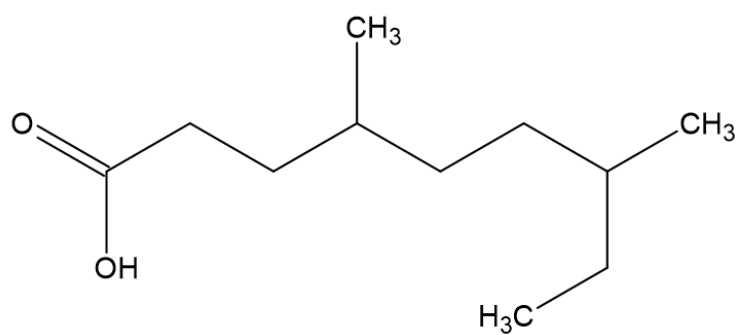
1. Find the longest carbon chain and drop the "e" from the parent alkane name and add "-oic acid" to the suffix (ex. pentanoic acid)
2. Carbon 1 will always be the carbon that is part of the carboxyl group

Example: Name the following molecule



1.9.2

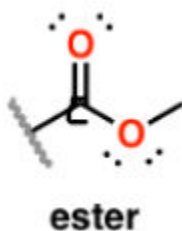
Name the following molecule:



1.10 Esters

1.10.1

Naming and Properties of Esters



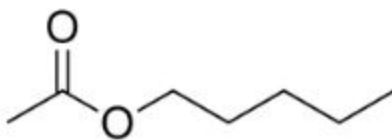
- An ester group is made up of a carbonyl group (C=O) bonded to an oxygen atom

Properties:

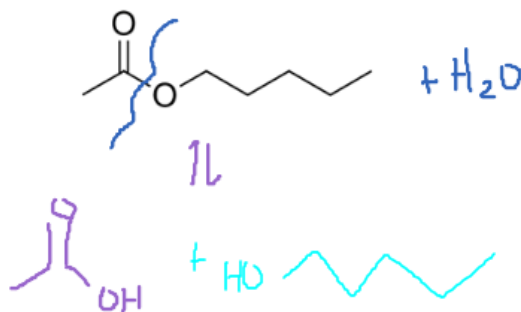
- Esters have the carbonyl group that is polar but doesn't have the OH group (like carboxylic acid and alcohols have)
- As a result, esters can't form hydrogen bonds with each other and have lower boiling points than carboxylic acids and alcohols

Naming:

Example: Name the following molecule



Carboxylic acid + alcohol $\rightarrow$ ester + water



 WIZE CONCEPT

To name an ester:

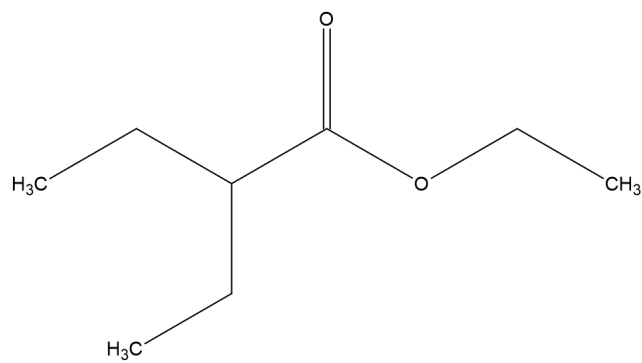
1. The first part of the name comes from the alcohol. Ignore the O atom and name the alkyl chain (ex. methyl, ethyl, propyl, etc)
2. Find the part of the ester that came from the carboxylic acid. Count the longest chain of carbon atoms. Take the parent chain name, drop the "e" and add "oate" to the ending (ex. butanoate).
 2. a) If there are substituents on the part that came from the carboxylic acid, name them before the main chain with the carboxylic acid.

The name of the ester will be in this form:

(alkyl chain that came from alcohol) SPACE (#)-(substituent on part from carboxylic acid)
(name of parent chain that came from the carboxylic acid with the oate ending)

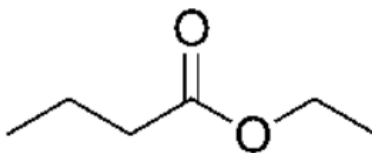
Name the ester above:

Example: Name the following molecule



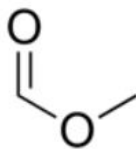
1.10.2

Name the following molecule:



1.10.3

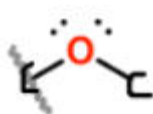
Name the following molecule:



1.11 Ethers

1.11.1

Naming and Properties of Ethers



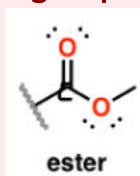
ether

! WATCH OUT!

Don't confuse ethers with esters!

Ethers has an oxygen atom single bonded to 2 carbon atoms.

Esters were made of a carboxylic acid group combining with an alcohol and releasing water (results in carbonyl group (C=O) bound to O group).



ester

Properties:

- Have an electronegative O atom, making them more **polar** than hydrocarbons
- Have stronger intermolecular attractions than hydrocarbons, but weaker intermolecular attractions than alcohols (which can hydrogen bond with each other)
- Has **higher boiling points than analogous hydrocarbons, but lower boiling points than analogous alcohols**

Naming:

 WIZE CONCEPT

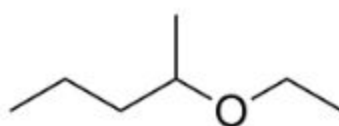
There are two chains connected to the oxygen atom...

1. Find the longer hydrocarbon group connected to the O atom and number it starting at the end of the chain closest to the O atom.
2. The first part of the name comes from the shorter hydrocarbon chain bound to O. Indicate the number of the carbon atom that is bound to the O atom first and then name it with the prefix (ex. meth, eth, prop, but, etc) and add the word "oxy" to the end of it.
3. Look at the longer hydrocarbon chain again and indicate the position # and branch names of any branches as you normally would on the parent chain connected to the O atom.
4. Name the parent alkane as usual (ex. methane, ethane, propane, etc).

Final naming:

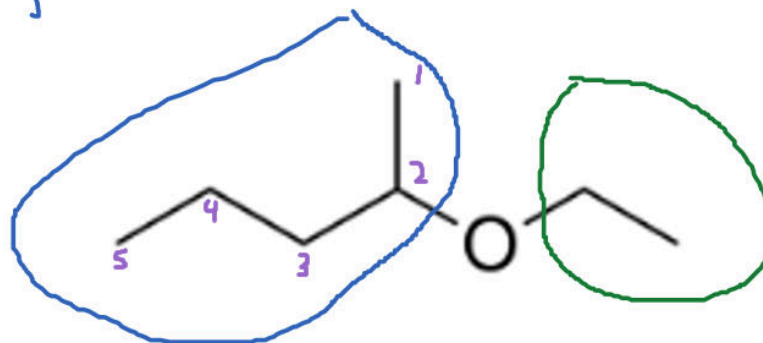
(# of C atom in long chain attached to O atom)-(Short hydrocarbon prefix name)(oxy)(#-branches on main chain)(name of parent alkane)

Example: Name the following molecule

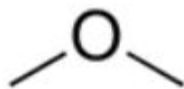


Longer chain

Shorter chain

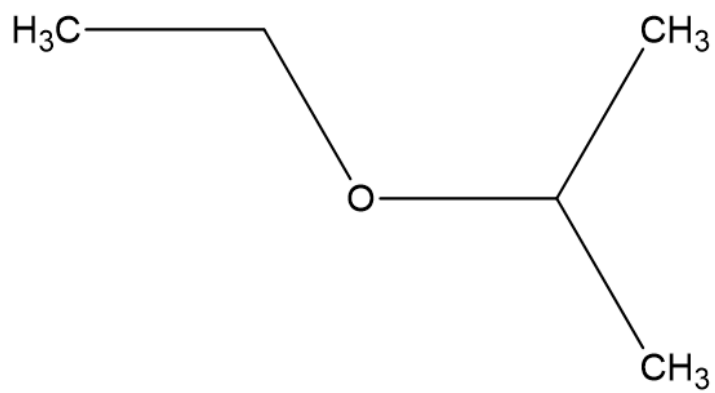


Example: Name the following molecule



1.11.2

Name the following molecule



1.11.3

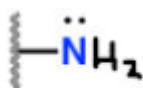
Draw the following molecule:

2-phenoxyheptane

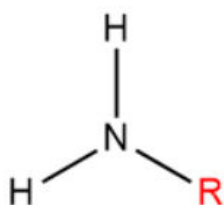
1.12 Amines

1.12.1

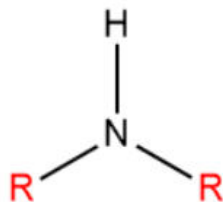
Naming and Properties of Amines



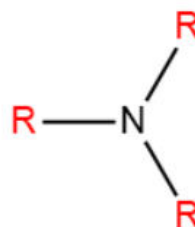
amine



Primary (1°)



Secondary (2°)



Tertiary (3°)

Properties:

- Amines have N-H and N-C bonds, both are **polar**
- The tertiary amine has no N-H bonds so cannot hydrogen bond with itself
- Since amines have polar bonds, they have **higher boiling points than hydrocarbons** of similar size
- Compared to alcohols (-OH) O is more electronegative so **OH bonds are more polar than NH bonds**. This is why **amines boil at lower temperatures than alcohols**

Naming:

 WIZE CONCEPT

1. Find the longest chain bonded to the nitrogen atom and name the parent alkane. This is the root of the name. Drop the "e" so it becomes methan, propan, etc.
2. Number the carbon atoms in the longest chain starting with the side closest to the N atom.
3. Place the number of the carbon atom in the longest chain that is bonded to the N atom in front of the suffix "-amine" to indicate it's an amine.
4. For secondary and tertiary amines, name the small hydrocarbon chain(s) as well. These will be the prefix. Write N- in front of each group and order the groups alphabetically. N-ethyl-N-methylhexan-3-amine or N,N-diethylhexan-3-amine.

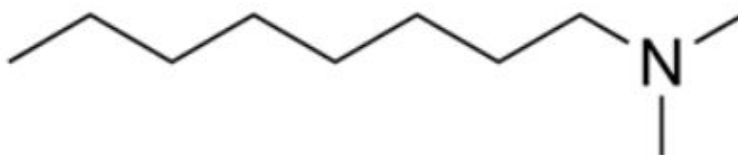
Name has the form:

N-(alkyl chain name of smaller chain)-N-(alkyl chain of other small chain)(root name of longest chain with "e" removed)-# of C atom in long chain N is bound to-amine.

Example: Name the following molecule

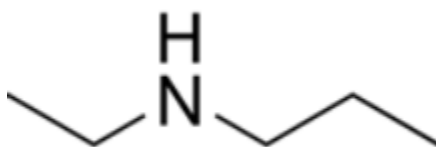


Example: Name the following molecule



1.12.2

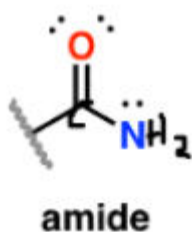
Name the following molecule



1.13 Amides

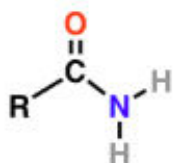
1.13.1

Naming and Properties of Amides



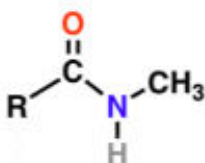
- The amide functional group has a **carbonyl group** (-C=O) bound to a **N atom**

Amides: count the number of carbons directly attached to the nitrogen



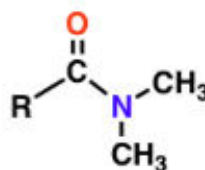
1 carbon
directly
attached

**Primary (1°)
amide**



2 carbons

**Secondary (2°)
amide**



3 carbons

**Tertiary (3°)
amide**

Properties:

- Have the **polar carbonyl group** and primary and secondary amides also have **N-H bonds** giving them the ability to **hydrogen bond with themselves**
- More N-H bonds means that amide will have a higher boiling point (more hydrogen bonds)
- Boiling points are higher than other hydrocarbon derivatives** of the same size

Naming:

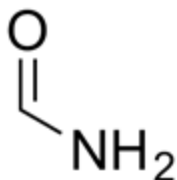
 WIZE CONCEPT

1. Determine the longest hydrocarbon chain that includes the carbonyl carbon and name it but drop the "e" ending (ex. methan, ethan, propan, etc)
2. Add "amide" to the ending.
3. For prefixes, first look to see if there are any other alkyl chains on the N atom. If there are, name them first (ex. N-ethyl-N-methylpropanamide or N,N-dimethylpropanamide)
4. Now number the main hydrocarbon chain with the carbonyl carbon so that the carbonyl carbon is #1.
5. Indicate the position of the alkyl chains on the main chain after writing out alkyl chains attached to N:

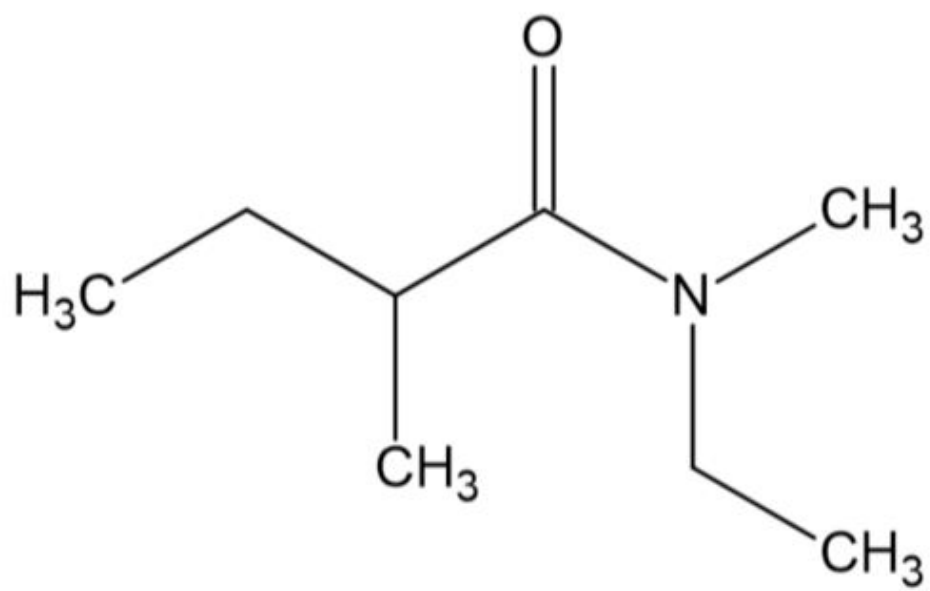
Format: N-(alkyl chain attached to N)-N-(alkyl chain attached to N)-#-(alkyl chain attached to parent chain)(name of parent chain without e)(amide)

ex. N-methyl-2-methylpropanamide

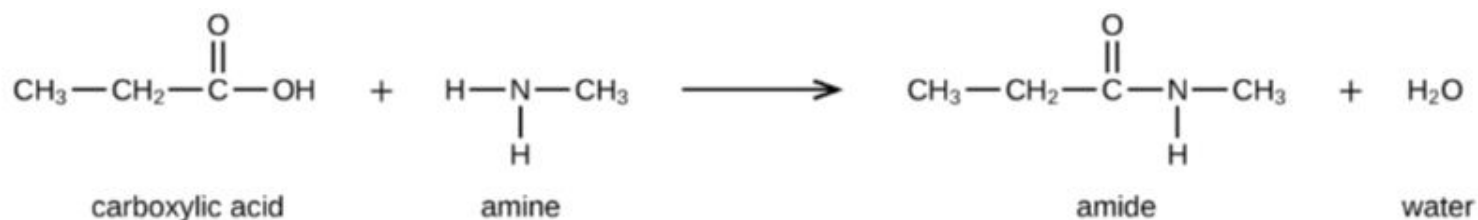
Example: Name the following molecule



Example: Name the following molecule



Forming Amides



Name the amide in the reaction above:

The first part of the name comes from the **amine**

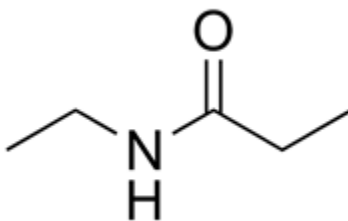
The second part of the name comes from the **carboxylic acid**

Review: What was the functional group that we made with a carboxylic acid + alcohol? _____

Draw this functional group out:

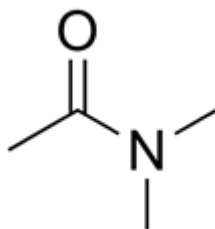
1.13.3

Name the following molecule:



1.13.4

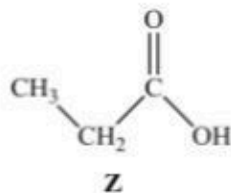
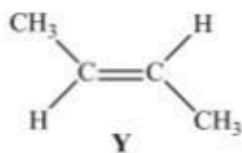
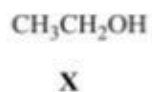
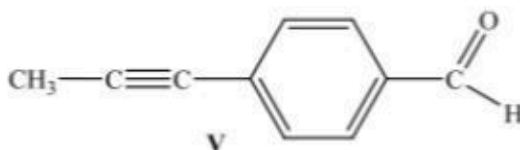
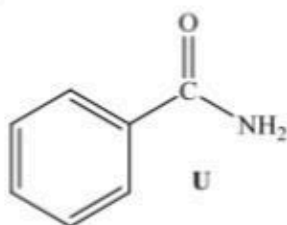
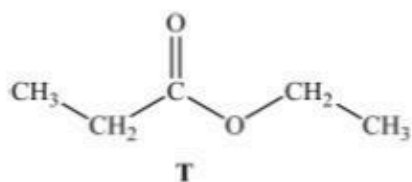
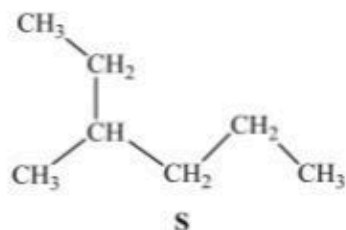
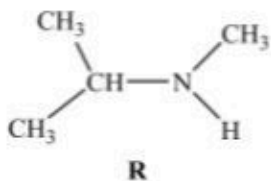
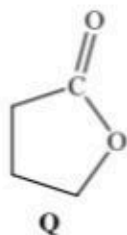
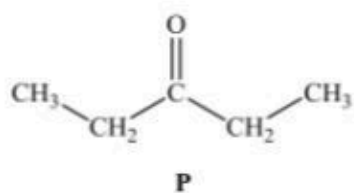
Name the following molecule:



1.14 Summary and Checklist

1.14.1

Match the correct letters Ex. P with the correct functional group (ex. alcohol). Try your best!



-
- A.** ketone
 - B.** carboxylic acid
 - C.** alkane
 - D.** aldehyde and alkyne
 - E.** ester
 - F.** alkene
 - G.** amide
 - H.** alcohol
 - I.** amine
 - J.** ether

P

Q & T

R

S

U

V

W

X



Y



Z

Summary of Properties (Polarity)

This is not usually tested on but I still recommend going over this for **understanding .

In order of decreasing polarity:

Amide > Carboxylic Acid > Alcohol > Ketone ~ Aldehyde > Amine > Ester > Ether > Alkane

i WIZE TIP

For exams, it can be helpful to have a general idea of which functional groups are more polar and which are less polar!

See examples below

- Which is more likely to dissolve in water? A carboxylic acid or ether? _____
- Which is more non-polar? A ketone or an alkane? _____

Organic Chemistry (Part 1: Functional Groups and Naming) Summary Checklist

After going through this chapter you should be familiar with each of the concepts below!

- Drawing structural/molecular formulas (complete and condensed) for organic compounds
- Writing molecular formulas (full and expanded for organic compounds)
- Naming hydrocarbons (aliphatic and aromatic)
- Identifying and naming isomers (constitutional and stereoisomers) for organic compounds
- Providing IUPAC names, drawing structures, identifying/naming functional groups for compounds for all organic families discussed (ex. alkanes, alkenes, alcohols, carboxylic acids, etc).